Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.05	20.23	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.00	2.02	+0.9%
power (W), mean	625	643	+2.8%
power (W), peak	646	681	+5.3%
energy (Wh)	209	217	+3.8%
energy, parked workers still on at idle power (Wh)	209	217	+3.8%
CPU used (cores), mean	0.784	1.24	+58.5%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.033	0.052	+58.5%
energy per core-hour (Wh)	797	517	-35.1%
node-hours per core-hour	7.65	4.83	-36.9%
pending pods, pod-minutes	0.533	0.533	+0.0%
pending pods, peak	2.00	2.00	+0.0%
HPA replicas, mean	8.59	9.43	+9.8%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.783	0.517	-34.0%
requests timed	1,130	1,100	-2.7%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	62.05	95.27	+53.5%
response time (ms), median	42.90	81.95	+91.0%
response time (ms), 95th percentile	135	152	+12.8%
response time (ms), 99th percentile	206	174	-15.5%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	8.4804	5.4570	-35.7%	[-4.8710, -1.1385]	GAIN
kwh_per_core_hour	0.8795	0.5797	-34.1%	[-0.4860, -0.1200]	GAIN
utilisation	0.0324	0.0509	+56.9%	[+0.0066, +0.0314]	GAIN
pending_pod_minutes_per_hour	1.4634	1.5238	+4.1%	[-4.3902, +4.5714]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.2322	1.5238	-31.7%	[-3.6722, +2.2857]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.358 0.392 0.42 0.445 0.468 0.487 0.504 0.52 0.534 0.546 0.556 0.562 0.566 0.571 0.576 0.581 0.585 0.589 0.59 0.591
- nodes decided per minute: 6 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	613	633	4.00	5.00	0.000	0.500	0.435	0.999

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	618	625	4.50	8.00	0.000	0.000	0.581	0.779
2	6.00	6.00	612	623	5.00	8.00	0.000	0.000	0.430	0.729
3	6.00	6.00	616	631	5.00	8.00	0.500	0.000	0.543	0.927
4	6.00	6.00	630	648	8.00	10.00	0.000	0.000	0.936	1.40
5	6.00	6.00	628	649	8.00	10.00	0.000	0.000	0.853	1.41
6	6.00	6.00	629	647	8.00	10.00	0.000	0.000	0.894	1.37
7	6.00	6.00	644	672	10.00	10.00	0.000	0.000	1.31	2.02
8	6.00	6.00	642	675	10.00	10.00	0.000	0.000	1.28	2.09
9	6.00	6.00	641	676	10.00	10.00	0.000	0.000	1.23	2.12
10	6.00	6.00	637	669	10.00	10.00	0.000	0.000	1.09	1.96
11	6.00	6.00	615	626	10.00	10.00	0.000	0.000	0.514	0.809
12	6.00	6.00	615	620	10.00	10.00	0.000	0.000	0.516	0.651
13	6.00	6.00	619	634	10.00	10.00	0.000	0.000	0.626	0.910
14	6.00	6.00	632	652	10.00	10.00	0.000	0.000	0.945	1.49
15	6.00	6.00	631	648	10.00	10.00	0.000	0.000	0.968	1.36
16	6.00	6.00	629	649	10.00	10.00	0.000	0.000	0.888	1.42
17	6.00	6.00	621	639	10.00	10.00	0.000	0.000	0.685	1.18
18	6.00	6.00	617	624	10.00	10.00	0.000	0.000	0.542	0.738
19	6.00	6.00	615	624	10.00	10.00	0.000	0.000	0.512	0.723

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.